Task 3 – Answering questions – General

CONTENTS

Causal mapping produces models you can query to answer questions

20 Sep 2025

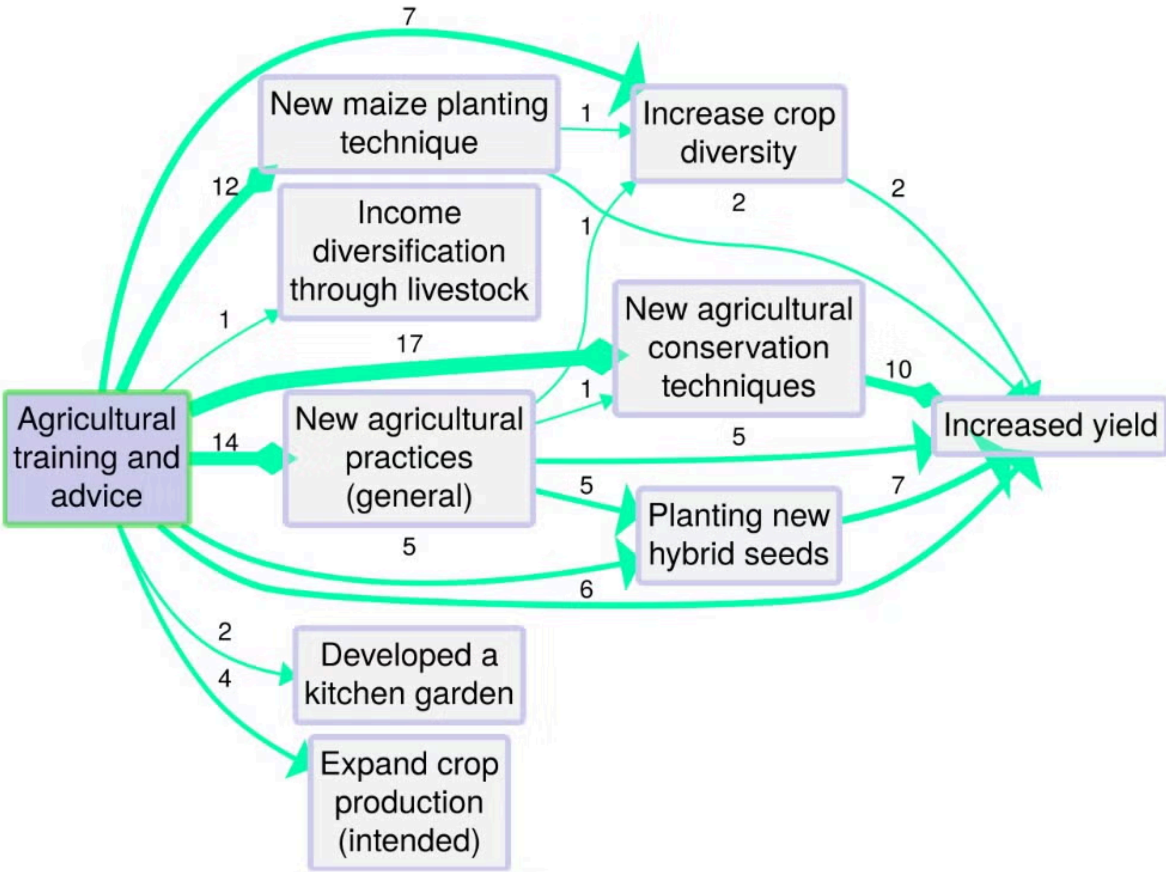
The fundamental output of causal mapping is a database of causal links. If there are not too many links, this database can be visualised "as-is" in the form of a causal map or network. But usually there are too many links for this to be very useful, so we apply filters.

By applying filters and other algorithms, a causal map can be queried in different ways to answer different questions, for example to simplify it, to trace specific causal paths, to identify significantly different sub-maps for different groups of sources, etc.

As explained on the [Causal Mapping website](#):

"*A global causal map resulting from a research project can contain a large number of links and causal factors. By applying filters and other algorithms, a causal map can be queried in different ways to answer different questions, for example to simplify it, to trace specific causal paths, to identify significantly different sub-maps for different groups of sources, etc.*"

The figure below shows a map from the application Causal Map, showing coded causal statements for a project that provided farmers with agricultural training and advice in order to increase crop yields. The map has been filtered to show only outcomes downstream of the influence factor 'Agricultural training and advice'. Numbers shown indicate how many times the links were mentioned across all interviews.



Source: BDSR, 2021, p 4

Outputs of QDA

2 Oct 2025

The logic of QDA: you’ve done your analysis, now what?

The result of qualitative data analysis can be understood as, at least, some kind of qualitative theory or model at least of the sources’ beliefs, with at least some possibility of generalising beyond them. But it can be hard to know what to do with the results of an emergent qualitative text analysis. There is no clear decision procedure: we can ask the author, and the answer is: some explanation, i.e. more text. In more reproducible approaches we do get some more structured outputs such as tables of frequencies. Some authors such as Mayring see these kinds of outputs as an important analysis result. QDA software is often used to capture and structure and even make inferences with these kinds of outputs.

In the logic of (non-causal) reproducible QDA, we can do things like this:

- - count occurrences of concepts, and use ordinary arithmetic to report eg which of two concepts was more common
- - count co-occurrences of two concepts, and construct measures like association between concepts, and more generally combine and query occurrences with boolean logic
- - create case/code matrices
- - report relationships between sources and concepts, for example to compare codings of one concept for different genders
- - reason about concepts, for example to deduce that an occurrence of "lion" is also an occurrence of "mammal", either relying on our implicit understanding of the concepts or through the explicit declaration of a parent-child relationship.

Of course frequency statistics are notoriously unstable, because they depend on our decisions about granularity and chunking. If I have a codebook which has 100 different codes for cats and only 1 code for dogs, we may conclude that dogs were mentioned in the text more often than any other animal-concept even if cat-concepts were mentioned more often in combination. This is one reason why reasoning with these kinds of outputs can never be merely automated. There always has to be a “human in the loop”.

Nevertheless the point is that we can understand the output of QDA coding as some proportion of "more text", which itself needs to be interpreted by humans, and a complementary proportion of machine-readable, structured output which can be used to ask and answer questions (Which are the overarching themes? How much does climate anxiety come up as a theme? Who mentions it most?) at least somewhat independently of human guidance.

QDA logic can also be extended beyond the simple logic of frequencies and occurrences to apply (special kinds of) codes which have additional explicit rules associated with them, such as code weighting (as for example in MaxQDA). This means we can for example apply codes like "somewhat happy" or 'very unhappy' which enable us to say that the expression of happiness in one case is stronger than the other, or (if we also allow coding for time) that happiness increases or decreases over time. These extra deductions we can make come free with the (implicit or explicit) underlying ordinal logic of comparison of intensity.

QDA without coding

Coding does not have to be central to qualitative data analysis (Morgan 2025; Nguyen-Trung & Nguyen 2025). ...

References

Morgan (2025). *Query-Based Analysis: A Strategy for Analyzing Qualitative Data Using ChatGPT*. https://doi.org/10.1177/10497323251321712.

Nguyen-Trung, & Nguyen (2025). *Narrative-Integrated Thematic Analysis (NITA): AI-Supported Theme Generation Without Coding*. https://doi.org/10.31219/osf.io/7zs9c_v1.

The fundamental property of causal maps is transitivity

9 Oct 2025

How does causal inference work in a causal *network*?

When is a pathway not just a link?

The logic around how links might combine into pathways and what that means for evaluation, that's the most exciting part. e.g. how might this intervention influence an outcome which might be multiple steps downstream of it?

From

> a -->b

and

> b --> c

what can we conclude about

> a --> c. ?

For example, if the relation --> means "causes", when and under what circumstances can we conclude that a causes c?

Once we know the inference rules for a network, in particular the transitivity rule, we can infer all kinds of useful things about it.

There is a whole library of thinking about causal reasoning within a statistical or probablistic network.

There is less written about qualitative causality within a qualitative causal network.

But our problem is harder again: to reason with what we call a causal map, where the links are about **beliefs about** or **evidence for** a causal connection.

[We can reason about causal maps using a logic of evidence]

Ways to causal inference

9 Oct 2025

There are different ways to set up a table like this. Some of these are more like methods, some are more like frameworks. Some overlap. Most do not exist only or even primarily to conduct causal inference.

Commonalities

Almost all these meta-models presuppose that causal knowledge can be captured in chunks, and these chunks can be joined together in chains / networks in order to consider indirect effects, ie if X causes Y and Y causes Z, what can we say about the causal effect of X on Z? Everything doesn't depend on *everything* else: in a large causal network, to make causal inferences about Z the only causal factors we need to consider are those for which there is a causal chain leading to Z, and the only direct effects on Z are the factors which have arrows directly leading to Z.

"Portability" means that the knowledge that Xs can influence Ys can be applied in multiple contexts, not just on one particular occasion.

Traditionally, more qualitative approaches like QCA are less keen on portability.

Some of the models eg SEM are particularly interested in calculating indirect effects in chains and networks. In others eg QCA the concept of a causal chain is less clear, and there is little portability / generalisibility.

Table

	Notes	Prerequisites	Inference	Metaphor	
Ordinary reasoning		I already know that Xs are likely to cause Ys, and this X happened and Y followed it in just the way you'd expect. I don't exactly have a ready-made theory about X's and Y's but I have similar theories about things similar to Xs and Ys and some other knowledge which suggest that the these similar	It is quite likely that X causally influenced Y.	Both the foundation on which all other methods stand and the cement which holds them together	

Ordinary reasoning: Modus Operandi		The signatures of all the other possible explanations of Y were absent and the signature of X was present. A signature is evidence of a chain from X to Y or other evidence that X was active. Relies on pre-existing causal knowledge Argues also that counterfactuals may be irrelevant; the patient might also have died if they hadn't had the heart attack, but the heart attack was the direct physical cause.			Scriven
Ordinary reasoning: Multiple lines and levels of evidence (MLLE)					
Ordinary reasoning: Causal mapping		Not really a method of causal inference at all, but a way to organise medium or large datasets of pre-existing (possibly conflicting or overlapping) causal claims or inferences made by a set of sources. Says nothing about portability because it relies			

		on its sources to decide on what			
Successionist		- Many observations that Ys follow Xs	May in practice be used to support but not warrant the conclusion that the Xs caused the Ys. Might be used in evaluation as a hoop test.		Hume
Classical Statistical		Not possible			
Regression discontinuity					
Counterfactual		Can be used for **disproving** a causal inference: if sometimes Xs appear but Ys do not appear, this X cannot be the cause of this Y.	X caused Y		
DAGs / SEM		Level 3: A model in which the effect of X on Y is significant Substantive theory			Pearl
Configurational; QCA, fQCA		A dataset showing the co-occurrence and non-occurrence of X1, X2, X3 … and Y A case in which X1, X2, X3 etc and Y occur/don't occur in a pattern corresponding to the dataset Substantive considerations backing up the dataset Possible	This combination of Xs may have caused Y in this case.	Causal package	Ragin

		consolidation of the information into more parsimonious structures. Usually not used to establish general causal laws and then make specific causal inferences but only to make causal inferences within a dataset. The inference that X1, X2, X3 actually *cause* Y rather than just being associated with it is not based only on the configuration but has to be provided by other substantive knowledge, so perhaps QCA is not strictly a method of causal inference.			
INUS		Special case of the above in which X1, X2, X3 … are necessary parts of a package P which is sufficient for Y, i.e. (X1 & X2 & X3…) is sufficient for Y.	X1 was an INUS-cause of Y (and so was X2 etc)	Causal package	
Realist		Substantive knowledge that a mechanism M in a context C, when triggered (X) causes Y. M and C were present, X was	X was a cause of Y	A machine or mechanism	

		activated, Y happened.			
Process oriented; Process tracing					
Physical causality		“people’s “operative reasons” for doing what they do are the physical actions.” “A design whose purpose is to determine impact will be considered qualitative if it relies on something other than evidence for the counterfactual to make a causal inference. It is qualitative first in the positive sense that it rests on demonstrating a quality-in the present treatment the quality will be a physical connection in the natural world between the proposed cause and effect-and second in the negative sense that it is not quantitative, that is, it does not rely on a treatment variable, or on comparing what is with an estimate of what would otherwise have been.”		Touching	Mohr
Dispositional					

Radical systems eg Quinn Patton?		Not possible			

Epistemic logic does not help us with reasoning about causal maps

9 Oct 2025

(An example of kind-of qualitative causal logic, with a focus on groups: Castellani et al. (2025)])

> From (Powell et al. 2024)

Seen as models of the world, causal maps, like systems maps, are fallible but useful: We can use inference rules (which are explicitly set out in FCMs, SDs, BBNs and CLDs and are implicit in other related approaches), and in particular, transitivity rules, to make deductions about the world.

There are at least three problems of transitivity which we need to think about

1. Given that A influences B and B influences C, does A influence C?
2. Given that P believes that A influences B and P believes that B influences C, does P believe that A influence C?
3. Given that someone believes that A influences B and someone else P believes that B influences C, does someone (who? we? the people?) believe that A influence C?

So if A causes B and B causes C, causal logic might tell us the answer to 1) under what circumstances A causes C.

Seen as models of individuals' causal beliefs, we can arguably use analogous rules to make deductions about what individuals believe, or ought to believe, given what else they believe.

There is a thing called epistemic logic which is a strange shadow of causal logic. Can it help us answer 2 and 3?

But epistemic logic is a strange thing.

If a person P believes that A causes B and B causes C, epistemic logic tells us what P believes about A causing C *if they were a rational person*. Whereas, facts about what people actually do believe is a branch of psychology.

In the last decades, thinkers like Daniel Kahneman have shown that in this sense, humans are so far from rational that it does not make sense even to start off with a rationality assumption and then add some corrections.

It would be great to use causal maps to infer, given a bunch of information about different people's causal beliefs, what they believe about *other* causal connections. That would be really useful. But it is hard.

There is a much easier way to reason with causal maps which is also vital for evaluators: to reason about **evidence**.

[We can reason about causal maps using a logic of evidence]

References

Castellani, Schimpf, Wistow, Caden, Agarwal, & Barbrook-Johnson (2025). *Case-Based Systems Mapping: Advancing a Multimethod Approach to Social Complexity*. Routledge. https://doi.org/10.1080/13645579.2025.2564177.

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

We can reason about causal maps using a logic of evidence

20 Sep 2025

> From (Powell et al. 2024)

Evaluators can break the Janus dilemma and make the best use of causal maps in evaluation by considering causal maps not primarily as models of either beliefs or facts but as repositories of causal evidence. We can use more-or-less explicit rules of deduction, not to make inferences about beliefs, nor directly about the world, but to organise evidence: to ask and answer questions such as:

• Is there any evidence that X influences Z?

• . . . directly, or indirectly?

• . . . if so, how much?

• Is there more or less evidence for any path from X to Z compared to any path from W to Z?

• How many sources mentioned a path from X to Z?

• . . . of these, how many sources were reliable?

We also argue that this is a good way of understanding what evaluators are already doing: gathering and assembling data from different sources about causal connections in order to weigh up the evidence for pathways of particular interest, like the pathways from an intervention to an outcome.

References

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

Causal maps are knowledge graphs, but with wings

26 Aug 2025

What is a Knowledge Graph? 🧠

- A knowledge graph is like a **giant mind map for a computer**. It stores information not as text in a document, but as a network of interconnected facts.
- It's built from two main things: **entities** (the "nodes," representing real-world objects, people, or concepts like "Paris" or "Photosynthesis") and **relationships** (the "edges," describing how these entities are connected, like "is the capital of" or "is a process in").
- A single fact has three parts: **(Subject) --- [Relationship] ---> (Object)**. For example: `(Marie Curie) --- [discovered] ---> (Radium)`.
- Why are knowledge graphs specially useful in the age of AI? 💡
- **They create structure from chaos.** AI can read through millions of pages of unstructured text (like news articles or scientific papers) and pull out these factual triplets. This turns a messy sea of words into an organized, queryable database of knowledge.
- **They enable smarter searching and reasoning.** Instead of just searching for keywords, you can ask complex questions that require understanding the relationships between things. For example, "Which scientists who won a Nobel Prize also discovered an element?" A computer can navigate the graph's connections to find the answer.
- **They provide essential context.** A knowledge graph helps an AI understand that "Apple" in a tech article is a company linked to "Steve Jobs," not the fruit. By looking at its connections, the AI gets the right context, which is crucial for accurate understanding and analysis.

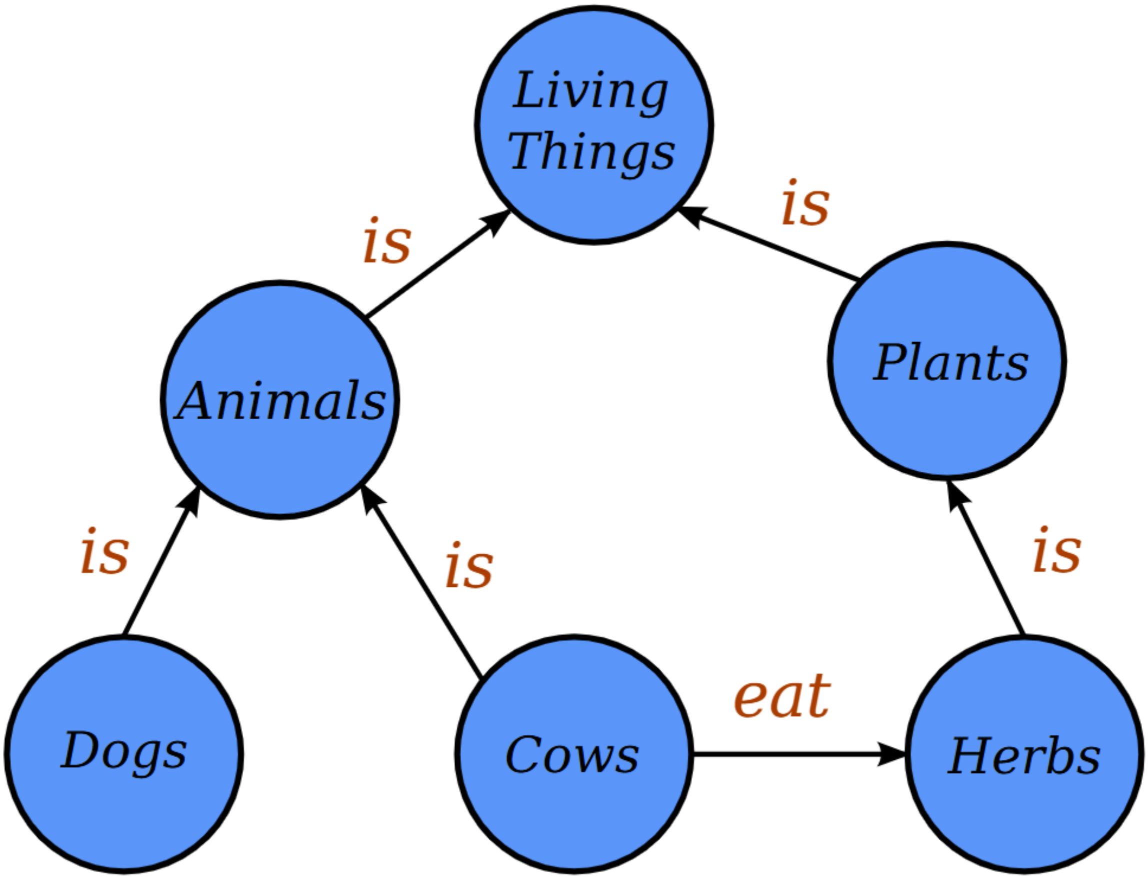


Image from Wikipedia by Jayarathina - Own work, CC BY-SA 4.0.

Why are Knowledge Graphs (KGs) so useful?

A major benefit of KGs is we can then apply network logic like transitivity rules to answer meaningful questions. For example, if the relation is "works in the same company as", then if we know

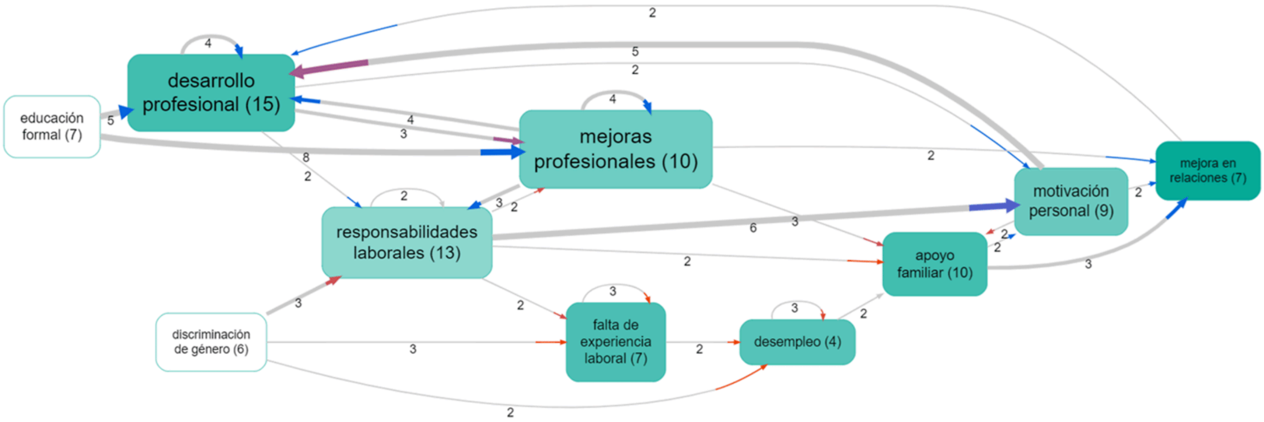
- **A is related to B**
- and
- **B is related to C,**
- then we can conclude
- **A is related to C (A is in the same company as C).**

Challenges with general-purpose knowledge graphs

The trick in **constructing** knowledge graphs is to know what relationship(s) to look for. "belongs to? " "is capital of?" "challenges/undermines?" This can be very difficult to decide. on the fly.

Using network logic to answer queries can be difficult where each different type of relationship may have its own logic. It can be very tricky (though potentially rewarding and useful) to design custom queries to answer specific questions.

Causal mapping gives knowledge graphs wings



A causal map is just a knowledge graph in which there is only one kind of relation: "causes" or "influences". This means:

- It is **much easier to scan and process text data** as we already know what we are looking for.
- We focus on primarily on **exactly the kind of information which is useful for monitoring and evaluation**: what influences what?
- We can **make use of pre-existing logic and queries to help answer common evaluation questions** almost "out of the box". Here we have a whole presentation on *Questions you can answer with causal mapping* which gives plenty of suggestions.

Can only doing causal mapping answer all the questions you might want to ask about a text? Of course not. But it can help answer a lot of the most interesting and important ones.

What about social network analysis?

Yes, social networks can also be constructed as knowledge graphs with just one (or a small number of) relationships, such as "works with".

So can we use causal mapping tools to construct general network graphs?

You might ask if the reverse is also true: can you use causal mapping software like Causal Map to also do your AI-supported knowledge graphing for you? The answer is yes!

You can code any type of link, not just causal, and you can also guide the AI to do this too.

[Social Network Analysis -- SNA]

The product of (causal) qualitative coding can be a model you can query

The result of qualitative coding of texts: is it a model?

Every type of Qualitative Data Analysis (QDA) is a bit different....

1. The most important result of coding might be a **deeper understanding of the text**. Something the researchers have in their heads. Perhaps something shared within a group. They can answer new questions about the data and write summaries of it from a new angle. In a sense you could say they have more or less formally developed a **model** of the data or even a theory about it, or, in the sense of Grounded Theory, a theory which goes beyond that data but is partly inspired by it. If you yourself read some of these different outputs and engage with them, maybe you yourself can start to build such a model in your head.

2. Apart from the intangible products in researchers' heads there are also **tangible products such as research reports**. Different schools of QDA give different importance to tangible as opposed to intangible outputs.

3. Thirdly, there may be **tabular outputs** such as tables of coding frequencies, cross-tabulations, etc. The definitive set of tables can be queried to answer additional questions like "how many women who mentioned working from home also see television as a an outdated medium". These kinds of tabular outputs can be quite useful to answer different questions.

So when you do an evaluation, what's the product? What do you get?

Obviously, you can think of the evaluation report, which might answer predetermined questions, but it may also include material that goes beyond the specific questions we were tasked with answering—for example, to address unanticipated issues or simply to describe or contextualize. Another important output is relational: hopefully, people have come together in a way that helps to expand learning and perhaps develop projects or relationships.

But today I want to talk about something different.

A statistical analysis not only answers questions but gives you the whole model -- can a qual analysis do that too?

When you do quantitative research, you might have specific research questions, but often one of the major outputs is a statistical model of the phenomenon. (There might be an effort to go beyond the data and hope that the model generalizes more widely, but that's not my focus here.) In the simplest case, the model might represent a suspected causal relationship—say, between the amount of screen time in the evening and difficulty falling asleep.

At the very least, the model allows us to look at a case in the dataset and say: on this day, this person looked at a screen for three hours and rated, let's say, a difficulty of four out of five falling asleep on some self-rating scale. Because we have the model, **we can explain that**: yes, this is quite a high level of difficulty, and it's explained at least partly by a high level of screen time, at least in this individual case. The model might also enable us to **make predictions**, like: people who, at least in this context, spend more than three hours on screens in the evening are, on average, going to experience a higher level of difficulty falling asleep.

A more sophisticated model will probably capture more variables, and many models—like directed acyclic graphs—link up these kinds of connections into a causal network, so you can explain or even predict how tweaking one variable will affect another variable downstream of it. Of course, there are other kinds of statistical models apart from causal models, but if you're reading this, you love causal models, don't you?

There's a relatively small but extremely well-funded section of evaluation activity based around this kind of statistical causal model, with randomized control trials (RCTs) as one facet. Ideally, an RCT is tasked not primarily with producing a model, but with answering a specific question, like: which is the better of these two interventions? Or: does this treatment work better than a placebo? But these are calculations conducted on the underlying model, which, from the point of view of workflow, is the major output of the work.

To generalise or not to generalise

There are two ways you might want to use that kind of model.

- One is to answer further questions about the same dataset—for example, to ask whether a particular subset (say, people over 70) differ in how screen time influences difficulty falling asleep, compared to other subgroups.
- If it's a sophisticated model, it might allow us to *generalize beyond* this specific context, perhaps by including more general variables like attention style or eye movement speed, which might help explain and predict behavior in other contexts.

Most quantitative researchers make a big deal about generalizing the model beyond the specific use case or context. In fact, the whole point of the study is normally to do that, and there's a whole armoury of tools, concepts, and arguments about how and under what conditions you can generalize a model to other people, other years, or even other countries with different kinds of screens, etc.

Can qualitative research do that?

The majority of evaluations aren't like that, although they might include a specific quantitative question somewhere in their terms of reference. In most cases, we think of the research output as a report in which the original (possibly modified) list of questions is answered, with additional narrative to summarize and link these sections.

Now, going beyond strictly quantitative paradigms, some evaluation projects will also include what we might call **qualitative modelling**. For example, if we're using QCA (Qualitative Comparative Analysis), apart from answering specific questions, we've likely also produced QCA-style tables, which could help us answer other questions beyond those we were actually tasked with. You might see those tables as annexes to the report. The same goes for causal loop diagrams and other techniques, which are essentially quantitative models but with a more restricted set of numbers. For example, in causal loop diagrams, we might model a variable like inflation with a number from -1 through 0 to +1, and do the same for variables like unemployment or military threats, building models of the relationships between these things using simplified numbers.

What I want to argue here is:

> any halfway decent evaluation, which at least implicitly gathers qualitative information about how things within the evaluation influence one another, can be considered as constructing a qualitative causal model. This is irrespective of the specific methods used—even if it doesn't include something explicitly called causal pathways analysis or causal mapping.

Qualitative models as products: theory, model, or something else?

In quantitative research, the idea of a "model" as a product is well established: you build a model, and then you can query it to answer new questions—even ones you didn't anticipate at the start. But what about qualitative research? Can the result of a qualitative analysis be a model in this sense, rather than just a set of answers to specific questions or a summary?

Of course (some) qualitative research produces models. Just don't call them that.

Some qualitative researchers do indeed conceptualize their results as models. For example, grounded theory often produces a theoretical model that explains the underlying processes or relationships within the data. These models can be revisited and "queried" to generate new insights beyond the initial research questions.

However, many qualitative researchers are more comfortable with the term "theory" rather than "model." "Theory" aligns more closely with the interpretive and conceptual nature of qualitative work, emphasizing explanation and understanding rather than prediction or parameterization. Still, the distinction is often more about language than substance: both models and theories can serve as frameworks for making sense of data and for generating new questions.

How are qualitative models used?

In qualitative research, especially in fields like grounded theory and narrative inquiry, the focus is less on "prediction" and more on generating understanding or insight. Researchers talk about "theorizing" from the data—developing concepts and frameworks that explain the phenomena under study. Once a theory or model is developed, it can be revisited, interrogated, and applied to new data or different contexts. This iterative process allows for continual refinement and deeper insight.

Importantly, a qualitative model or theory can also be used to answer new questions about the same dataset. For example, after developing a grounded theory, researchers (or others) can return to the theory and use it as a lens to interpret further cases, refine concepts, or generate new insights—without having to re-examine all the original data. This practice is sometimes referred to as "secondary analysis" or "theoretical application," where the theory or model functions as a standalone analytical tool.

In causal mapping, for instance, the model might consist of a network of causal links derived from qualitative data. Even if there are no quantitative parameters on the links, the model can still be queried: "Is there evidence for a causal pathway from A to B?" or "What are the main factors influencing outcome X?" This allows the model to be used flexibly, supporting both anticipated and unanticipated lines of inquiry.

Why does this matter?

Thinking of qualitative research outputs as models (or theories) that can be queried and reused has several advantages:

- **Transparency:** It makes explicit the structure of the findings and how they relate to the data.

- **Reusability:** Others can use the model/theory to answer new questions or apply it in new contexts.
- **Iterative learning:** The model can be refined and expanded as new data or perspectives emerge.
- **Bridging paradigms:** It helps bridge the gap between qualitative and quantitative traditions, showing that both can produce structured, interrogable outputs.

In summary, while qualitative researchers may prefer the language of "theory" over "model," the idea is the same: a well-constructed qualitative analysis can produce a framework that is more than just a set of answers—it is a model of the phenomenon, one that can be queried, shared, and built upon.

The transitivity trap

22 Aug 2025

> From (Powell et al. 2024)

[Granularity, generalisability and chunking are coding problems for causal mapping too]

Transitivity is perhaps the single most important challenge for causal mapping. Consider the following example. If source P [pig farmer] states 'I received cash grant compensation for pig diseases [G], so I had more cash [C]', and source W [wheat farmer] states 'I had more cash [C], so I bought more seeds [S]', can we then deduce that pig diseases lead to more cash which leads to more seed (G à C àS), and therefore G à S (there is evidence for an indirect effect of G on S, i.e. that cash grants for pig diseases lead to people buying more seeds)?

The answer is of course that we cannot because the first part only makes sense for pig farmers, and the second part only makes sense for wheat farmers. In general, from G à C (in context P) and C à S (in context W), we can only conclude that G à S in the intersection of the contexts P and W. Correctly making inferences about indirect effects is the key benefit but also the key challenge for any approach which uses causal diagrams or maps, including quantitative approaches (Bollen, 1987).

> For want of a nail the shoe was lost,
>
> For want of a shoe the horse was lost,
>
> For want of a horse the rider was lost,
>
> For want of a rider the battle war lost,
>
> For want of a battle the kingdom was lost,
>
> And all for the want of horseshoe nail.

(Thanks to Gary Goertz for remembering this one!)

Frog thinks: eating salad leads to health (less scurvy), and health (general fitness) leads to better sprinting ability, therefore if I eat this yummy lettuce – AARGH!

One of the key features of causal maps is that you can draw inferences, make deductions, from them. One of the most exciting is to be able to trace causal influences down a chain of causal links. BUT, when you are drawing conclusions from causal maps, beware of the transitivity trap:

from

> B → C

and

> C → E

we can only conclude

> B → E in the intersection of the contexts of 1 and 2

… and in general with any causal mapping, you'll never be sure that these two contexts do intersect. You actually have to look at each chain and think about it, and hope you've been told all the relevant facts.

For example:

If

> Source P [pig farmer]: I received cash grant compensation for pig diseases (G), so I had more cash (C)

and

> Source W [wheat farmer]: I had more cash (C), so I bought more seeds (S)

can we deduce

> G → C → H

and therefore

> G → S

(cash grants for pig diseases lead to people buying more seeds)?

No, we can’t, because the first part only makes sense for pig farmers and the second part only makes sense for wheat farmers.

There are thousands of different kinds of transitivity trap. It isn’t just a problem across subgroups of people. It can apply for example in different time frames.

If

> Child does well in year 13 (A) → Child has improved academic self-image (C)

and

> Child has improved academic self-image (C) → Child does better in year 9 (D)

can we deduce

> A → C → D

and therefore

> A → D

(child doing well in year 13 leads to child doing well in year 9)?

Of course not - even though these claims might be true of the same child. The problem arises as soon as we generalise one causal factor to apply to different contexts. We have to do this, to make useful knowledge. But there are always pitfalls too.

Not just a problem for causal mapping

This is also true, isn't it, of any synthetic research / literature review?

And in statistics, knowing the effects from B → C and C → E means you can calculate the indirect effect of B on E but not the direct effect. You have to have additional data just for that. This is one source of various so-called paradoxes in statistics.

Can we mitigate the trap with careful elicitation protocols?

Sometimes, we might know that all the information in one particular chain came from the same source, and all this information was explicitly given as a series of explanations of the factor which was initially in focus. But even here, we have to be careful. We might have to ask again, having reached the end of the chain, "did B really influence C which influenced D which influenced E? Was this all part of the same mechanism?" Are we sure we know exactly what we mean by this, and are we sure that our respondents do too?

In any case, part of the point of causal mapping is the synthetic surprises which we can discover by piecing together fragments of causal information which were *not* necessarily provided in this way. This is the situation every evaluator is in when piecing together information from, say, experts for Phase 1 and experts for Phase 2. We just always have to be aware of the transitivity trap.

Transitivity trap, or identity trap?

We can talk about the *identity trap* as more fundamental than the transitivity trap. It comes down to saying, how can you be sure that the way in which this factor is exemplified in one particular context is the same as the way that this similar seeming factor is exemplified in a different context: whether to use “the same” factor to code two different things.

References

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

